

M001 Week 8 — English Worksheet (Beginner)

Topic: Diamond Structure (Final Project) · **Course:** Pyramid Problems · **Level:** Beginner · **Time:** about 45 minutes

A diamond is two pyramids: the **bottom half grows** ($1 \rightarrow 3 \rightarrow 5$) and the **top half shrinks** ($3 \rightarrow 1$).

1 · Predict

Read the steps. Before you imagine the agent doing it, circle or write your answer.

```
# Bottom half
set size to 1
while size <= 5:
    [build layer]
    move up by 1
    set size to size + 2
```

The bottom half grows or shrinks? Circle one: grows · shrinks

Write the layer sizes in order:


```
# Top half
set size to 3
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

The top half grows or shrinks? Circle one: grows · shrinks

Why does it start at 3 and not 5? (Hint: the size-5 layer is already built.)

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The top half should start at **3**, so the widest layer is not built twice.

```
# clean
set size to 3
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

```
# buggy
set size to 5
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

What is wrong? Which layer appears twice?

Pair B — The bottom half must **move up** after each layer.

```
# clean
while size <= 5:
    [build layer]
    move up by 1
    set size to size + 2
```

```
# buggy
while size <= 5:
    [build layer]
    set size to size + 2
```

What is wrong?

3 · Fill the Gap

Here is the diamond skeleton. One starting number is missing. Fill it in using the word bank.

```
# Bottom half (grows 1 → 3 → 5)
set size to 1
while size <= 5:
    [build layer]
    move up by 1
    set size to size + 2

# Top half (shrinks 3 → 1)
set size to ____
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

Word bank: 3 · 5 · 1

Write the missing number:

4 · Tell Me What You Built 📷

Now build your **final project**: a diamond with a bottom half that grows **1** → **3** → **5** and a top half that shrinks **3** → **1**. When you finish, come back here.

Send a photo or video of your diamond, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

First, I built the bottom half by ...

The widest layer is ... blocks long.

The hardest part was ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) and walk the camera around your diamond. Talk like you are teaching a friend. Try to use these words: **diamond, bottom half, top half, widest.**

1. Walk around the diamond and show all 4 sides.
2. Say the layer sizes from bottom to top.
3. Say what you learned in this course.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk. This is the **last week** of Pyramid Problems — well done!